Steps:

1. Creating a VM
    a. First, I created a VM on VMWorkstation. I selected custom setup and kept most of the settings default. My method for adding the OS to the machine was to download the LTS version of Ubuntu to my desktop and using an ISO image, I copied it onto the machine. You can create an account for your machine and name your machine whatever you may like. I gave my vm 2gb of ram and 20 gb of disk (disk space is a few steps later), as this was the recommended minimum for Ubuntu.
    b. FOR THE TIME BEING, use a NAT network condition so you can download necessary things from the internet. The default options till the end is fine, using the recommended options is something I would recommend too.
    c. Resizing VM Disk space: I started with 20gb of disk, enough to comfortably install ubuntu and have some room to work. I wanted the vm to have 40 gb of disk space to have room for software and programs I would need to implement DNSSEC later. It is probably best to backup or take a snapshot of your vm before doing this step.
        i. Turn off the machine. Expand the disk in the VM settings, with expand disk to the desired size. Then, restart the machine and follow the commands below.
        ii. For commands, I ran:
            1. *lsblk* - I ran this to see where my root partition was, like /dev/sda1 or whichever says 20G. In the picture below, it is sda2, originally 20G, now 40G after resizing.

```
steve@steve-VMware-Virtual-Platform:~$ lsblk
NAME   MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
loop0    7:0    0     4K  1 loop /snap/bare/5
loop1    7:1    0  73.9M  1 loop /snap/core22/2045
loop2    7:2    0 251.6M  1 loop /snap/firefox/7672
loop3    7:3    0  73.9M  1 loop /snap/core22/2216
loop4    7:4    0 251.7M  1 loop /snap/firefox/7720
loop5    7:5    0  11.1M  1 loop /snap/firmware-updater/167
loop6    7:6    0  18.5M  1 loop /snap/firmware-updater/210
loop7    7:7    0   516M  1 loop /snap/gnome-42-2204/202
loop8    7:8    0 531.4M  1 loop /snap/gnome-42-2204/247
loop9    7:9    0  91.7M  1 loop /snap/gtk-common-themes/1535
loop10   7:10   0  49.3M  1 loop /snap/snapd/24792
loop11   7:11   0  10.8M  1 loop /snap/snap-store/1270
loop12   7:12   0   576K  1 loop /snap/snapd-desktop-integration/315
loop13   7:13   0  48.1M  1 loop /snap/snapd/25935
loop14   7:14   0   576K  1 loop /snap/snapd-desktop-integration/343
sda      8:0    0    40G  0 disk
├─sda1   8:1    0     1M  0 part
└─sda2   8:2    0    40G  0 part /
sr0     11:0    1   5.9G  0 rom  /media/steve/Ubuntu 24.04.3 LTS amd64
```

2. Next, ensure that you have cloud guest tools installed: *sudo apt-get update* and *sudo apt-get install cloud-guest-utils*
3. Next, expand the partition with *sudo growpart /dev/sda 1*, change the number to the partion number shown by lsblk.
4. Then run *sudo resize2fs /dev/sda<previous partion number here>* .
5. The run df -h to verify the new size

d. After reportioning, we can download some software before switching to a virtual network. For ubuntu, I downloaded BIND9 to set up my DNS Servers
   i. Command: *sudo apt install bind9* and *sudo apt install dnsutils*
      1. DNSutils provides helpful troubleshooting and debugging tools, and may not be installed with bind9

e. Also, install gcc to run c programs you might need later. If you would like to as well, install Wireshark on a VM as well to monitor and see the traffic you are sending.

f. VMTOOLS
   i. I tried numerous trouble shooting techniques to install VM Tools on the vm, but I ended up doing it through the command line instead of VMWare. I did not have administrative privileges on the machine and I think that is what prevented me from doing it through VMWare.
   ii. To install them, make sure you have an internet connection and run :
      1. *sudo apt-get install open-vm-tools-desktop*. Says yes when prompted, and once complete check the status with: *systemctl status vmtoolsd*. You should see "active" in green text.

g. At this point, cloning once for another machine to treat as a dns server is not the worst idea, I made the mistake of cloning a bit too early so I just had to repeat steps on multiple vms. It wouldn't hurt either to set up the virtual network and make sure your virtual network can be accessed by the virtual machine.

2. Setting Up the Virtual Network

a. Under the "Edit" Tab in the menu bar, there should be a Virtual Network Editor Option. Select this to begin setting up the network.

b. Select the "Add Network Option to create your new network.

c. Make the vmnet# one that is not taken yet. For me, 0, 1, and 8 were in use, so I used vmnet5. Select Host-only for the network type.

d. After selecting add and creating the network, you will be able to change specifics of the network. Enter the virtual machine settings and add a network adapter setting. Deselect DHCP service to distribute IP addresses to VMs. The IP addresses will be statically assigned by you. Connecting a host virtual adapter can also be deselected, it should not be needed to connect with the host on this network.
e. You can manually input a private IP address, like 192.168.x.0, or if you save, the editor will give you a random 192.168.x.0 IP address.

3. Cloning the VM
    a. Ensure the VM you intend to clone is powered off.
        i. Before cloning, ensuring that you have a Network Adapter that is connected to the vmnet you created in the previous step. To do this, navigate to the vm settings by right-clicking on the vm itself, clicking on the vm menu tab, or while it is turned off it can be seen while viewing the vm.
        ii. Under "Hardware", select add at the bottom of the screen. Select Network Adapter and finish.
        iii. Next, select the newly added Network Adapter, likely called Network Adapter 2 if you still have the original NAT connection.
        iv. Select Custom and your vmnet# should be available to choose from.
        v. At this stage, you shouldn't really need the NAT connection anymore, so you can remove it or deselect both of its connection options, "connected" and "connect at power on" so that it is easily available and can be turned on whenever, versus having to add it whenever you may need to access the internet. I would say it is best practice to delete it so that it cannot be defaulted to and you can ensure that your local network is working correctly.

4. Configuring the Machines IP
    a. On your Client Machine:
        i. Open a terminal and run the command:
            1. *ip a*. If you get an outpout like the one below, then you are likely also getting "Connection issues" popups. This is because the vm is recognizing your vmnet, but doesn't have an address yet. To set one, open up the network settings of your OS and navigate to IPv4 tab. Select manual for the method and then input an address. It'll match you vmnet, like 192.168.x.y, x being what you set or assigned your vmnet, and y being

whatever you like, between 1 and 255. I chose 130 to start, but its personal preference.

```
steve@steve-VMware-Virtual-Platform:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:7e:de:ef brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.107.130/24 brd 192.168.107.255 scope global noprefixroute ens33
       valid_lft forever preferred_lft forever
```

2. Then, for the subnet mask, input the same number that is listed on the vmnet.

ii. Also on the client machine, you should edit the netplan configuration. See the below photo to see the format, and replace the addresses field with your own client IP, name server field with your own name server ip, and search with your own domains. Ens36 is my leftover from when I added a NAT connection to download vmtools, and isn't necessary but it isn't hurting anything.

```
steve@steve-VMware-Virtual-Platform:~$ cat /etc/netplan/01-network-manager-all.yaml
network:
  version: 2
  ethernets:
    ens33:
      dhcp4: no
      addresses:
        - 192.168.107.130/24
      nameservers:
        addresses:
          - 192.168.107.129
        search:
          - lab
          - jmu.lab

    ens36:
      dhcp4: yes
```

b. On the Primary Server:

i. Open a terminal and run the command:

1. *ip a*. If you get a similar output as before, then follow the same steps as above to set the IP and Subnet.

2. However, for the DNS Server address, put the local 127.0.0.1, showing that this machine is a DNS Sever. Below should be the output after you add this to the settings:

```
steve@steve-VMware-Virtual-Platform:/etc/bind$ resolvectl status
Global
         Protocols: -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
  resolv.conf mode: stub
       DNS Servers: 127.0.0.1

Link 2 (ens33)
    Current Scopes: DNS
         Protocols: +DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
Current DNS Server: 127.0.0.1
       DNS Servers: 127.0.0.1
```

3. The below netplan may or may not be necessary, use it as a last resort if you are still unable to connect at the end.

```
steve@steve-VMware-Virtual-Platform:~$ cat /etc/netplan/01-network-manager-all.y
aml
network:
  version: 2
  ethernet:
    ens33:
      dhcp4: no
      addresses:
        - 192.168.107.129/24
    ens36:
      dhcp4: yes
```

c. Back on the Client Machine:

i. After setting the Primary Server IP, go back to the client and set the DNS in the same area of settings and put in the Primary Server address you just assigned.

ii. Try pinging the client vm from another machine. The primary server with a command like ping 192.168.107.130 to ping the client machine to check for connectivity.

iii. In a terminal, run the command *resolvectl status,* and you should see something like this below

```
steve@steve-VMware-Virtual-Platform:~$ resolvectl status
Global
         Protocols: -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
  resolv.conf mode: stub

Link 2 (ens33)
    Current Scopes: DNS
         Protocols: +DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
Current DNS Server: 192.168.107.129
       DNS Servers: 192.168.107.129
        DNS Domain: lab jmu.lab
```

d. On the Delegated Server
   i. Open a terminal and run:
      1. *Ip a*. For this machine, all you need to do is set the IP address and subnet mask as you did before. Try pinging both machines like you did before, and if the output is receiving packets from other machines, they are connected.

5. Setting up DNS on the VM
   a. Setting up a primary server.
      i. This server will be, as the name says, the primary server that will be used when querying DNS. I will use example.jmu.lab for setting up the DNS. A good name for this vm would be VM1.
      ii. First, you must add a zone. A zone in this instance would be "lab". To add a zone, you must enter */etc/bind/named.conf.local* and edit it.
         1. The syntax is seen below, and a db file is assigned to this zone.

```
//
// Do any local configuration here
//

zone "lab" {
        type master;
        file "/etc/bind/db.lab";
};
```

      iii. The db file does not automatically appear, hower making a copy of db.local into the name you gave it in the zone, mine being "db.lab". Now, nano into the newly created db file.
         1. With the db file, you see lines that start with an @. The first line contains a lot of info, the line with SOA (Start of Authority).

After the SOA, you need to edit the localhost part. For me, it would be lab. and root.lab.

2. Below that is the serial number, and this is an important part too. Every time you update this file, the serial number must be incremented too. It must only be incremented once if you make multiple changes before restarting bind9. The other numbers in the SOA aren't too important right now.
3. Next, you must edit the lines at the bottom. At a minimum, you need an example IN A 192.168.x.y, x being the network octet and y being an octet of your choosing.
4. To make the primary server an authoritative server, @ IN NS ns.<domain>.lab. line is needed.
5. and a ns IN A 192.168.x.y line, the x being your network ip and the y octet being whatever you like, but stay away from the ips you have reserved for the Client and DNS servers.
6. To make this server authoritative, the jmu.lab. and ns1.jmu.lab. lines are required, with the ns1.jmu.lab. line ending with the delegated server's IP address.
7. I haven't edited the top comment, and it might be misleading but that is the comment from db.local, please disregard.

```
steve@steve-VMware-Virtual-Platform:~$ cat /etc/bind/db.lab
;
; BIND data file for local loopback interface
;
$TTL    604800
@       IN      SOA     ns1.lab. root.lab. (
                              5         ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
;
@       IN      NS      ns1.lab.

ns1     IN      A       192.168.107.129

jmu.lab.        IN      NS      ns1.jmu.lab.
ns1.jmu.lab.    IN      A       192.168.107.128
bc      IN      A       192.168.107.1
computer        IN      A       192.168.107.2
```

8. When adding to the domain server, you will likely run into errors. I these are after changing named.conf file, run a command to check for those errors: named-checkconf. I you are having issues with a zone or named-checkconf return ok, the try named-checkzone <zone-name> /etc/bind/db.<zone-name>. If it tells you an error, correct it and and keep trying till you get an OK.

```
steve@steve-VMware-Virtual-Platform:/etc/bind$ named-checkzone jmu.lab db.jmu.la
b
zone jmu.lab/IN: loaded serial 6
OK
```

9. At this point, running a dig command at the primary server is a good idea. Using the domain you made above, run a dig command like so:
   a. *dig @<primary server ip> example.jmu.lab* (my example domain). As you can see below, the status area shows whether or not querying the server was successful. You should see NOERROR, but you might see SERVFAIL or NXDOMAIN

```
steve@steve-VMware-Virtual-Platform:~$ dig @192.168.107.129 computer.lab

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> @192.168.107.129 computer.lab
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 50660
;; flags: qr aa rd; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
;; WARNING: recursion requested but not available

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 77f6f6477cf686730100000069a5d81f095a8d9a76c0eb43 (good)
;; QUESTION SECTION:
;computer.lab.                  IN      A

;; ANSWER SECTION:
computer.lab.           86400   IN      A       192.168.107.2

;; Query time: 1 msec
;; SERVER: 192.168.107.129#53(192.168.107.129) (UDP)
;; WHEN: Mon Mar 02 13:34:07 EST 2026
;; MSG SIZE  rcvd: 85
```

10. SERVFAIL likely means there is an error in your network configuration, but running the check discussed above in 4.c.iv on the server could prove useful.

11. NXDOMAIN means that the was no domain found (Non-existent Domain), maybe there was a typo, or once again run the check discussed above in 4.a.iii.8 on the server could prove useful.

b. Setting up the Secondary Server
   i. For setting up the secondary server, you follow the same steps.
   ii. First edit /etc/bind/named.conf.local to be like this for the delegated zone, .jmu in this case:

```
//
// Do any local configuration here
//

zone "jmu.lab" {
        type master;
        file "/etc/bind/db.jmu.lab";
};
```

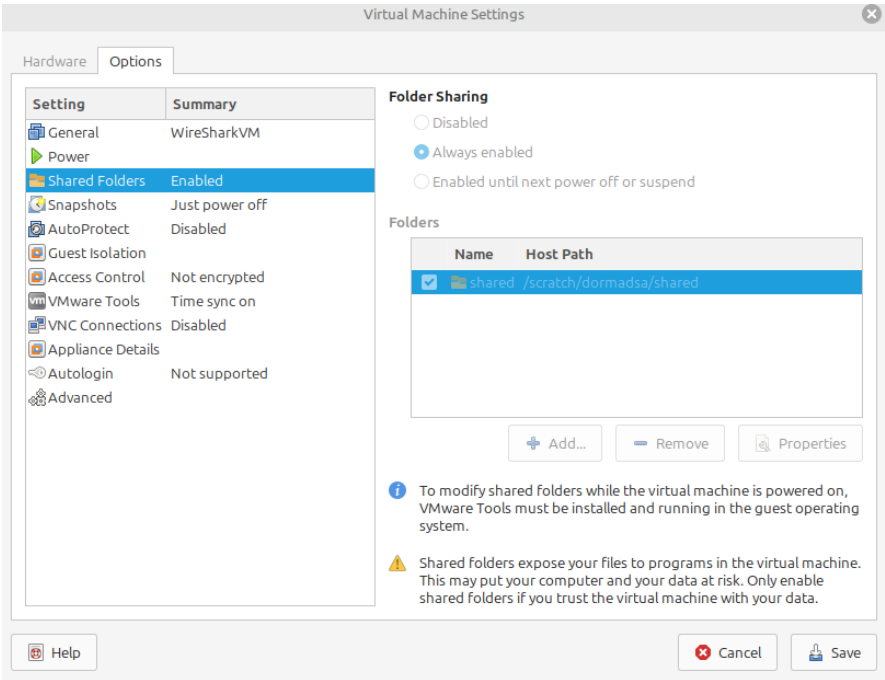
   iii. In this db file, the name space record points towards the server itself, and the other records are viable domains for <name>.jmu.lab.

```
steve@steve-VMware-Virtual-Platform:~$ cat /etc/bind/db.jmu.lab
;
; BIND data file for local loopback interface
;
$TTL    604800
@       IN      SOA     ns1.jmu root.jmu (
                              6         ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
;

@       IN      NS      ns1.jmu.

ns1     IN      A       192.168.107.128
example IN      A       192.168.107.50
shea    IN      A       192.168.107.69
john    IN      A       192.168.107.67
alley   IN      A       192.168.107.68
king    IN      A       192.168.107.66
main    IN      A       192.168.107.65
quad    IN      A       192.168.107.64
```

6. Creating a shared folder between the VM and Host machine

a. https://community.broadcom.com/vmware-cloud-foundation/communities/community-home/digestviewer/viewthread?GroupId=7171&MessageKey=e83f0009-629f-42a2-b0d0-a9f6ea56b3d0&CommunityKey=fb707ac3-9412-4fad-b7af-018f5da56d9f

b. First, enter the setting of the VM you would like to remove information from. Configure the settings while the VM is off, the Screenshow below was taken when the VM was on. As you can see below, switch over to the options tab, then go down to “Shared Folders”. Select Always enabled at the top, then “+ Add”. You can then set a name for the shared folder, as well as set the location you want it saved to on the host machine. I liked to keep it clean and make the name the same for both folders.



c. Then, start up the VM. Open a terminal and enter the command below. This will fuse the hgfs directory with the host, allowing you to then view your shared folder in the /mnt/hgfs directory. Now, you can copy or move files through the command line into this directory and they will be put into the location you specified in the previous step on the host machine.

```
steve@steve-VMware-Virtual-Platform:~$ sudo vmhgfs-fuse .host:/ /mnt/hgfs -o all
ow_other
steve@steve-VMware-Virtual-Platform:~$ ls /mnt/hgfs
shared
```

7. Capturing Traffic with Wireshark

a. I kept getting ntp.ubuntu network traffic, caused by my isolated network wanting to access the outside. To stop this, I ran sudo timedatectl set-ntp off to stop this bloat from coming through. This was sending hundreds of pings every couple of seconds, making it hard to filter out so I disabled it.

8. Configuring DNSSEC

a. https://documentation.ubuntu.com/server/how-to/networking/install-dnssec/

b. Keys are in /var/cache/bind

c. Getting started – Parent

i. Now that DNS is working, it is time to implement DNSSEC. As we will need to write to the disc, we have to move to another directory. I moved to /var/lib/bind/ to implement DNSSEC.

ii. Firstly, copy over db.lab file from /etc/bind/ to /var/lib/bind/. You will need to edit it like so: We are going to add some delegation code here too, we are saving a step for later.

```
$TTL    86400
lab.    IN      SOA     lab. root.lab. (
                              3         ; Serial
                          43200         ; Refresh
                            900         ; Retry
                        1814400         ; Expire
                           7200 )       ; minimum

lab.    IN      NS      ns.lab.
ns      IN      A       192.168.107.129

jmu     IN      NS      ns.jmu.lab.
ns.jmu  IN      A       192.168.107.128
; bc    IN      A       192.168.107.1
computer        IN      A       192.168.107.2
```

iii. Still on the parent machine, nano into /etc/bind/named.conf.options.

1. Outside of the options area, add a dnssec-policy like the one below. You need both a KSK and ZSK. The algorithm RSASHA256 can be swapped out for your algorithm of choice.

```
dnssec-policy rsa-policy {
        keys {
                ksk lifetime 365d algorithm RSASHA256 2048;
                zsk lifetime 60d algorithm RSASHA256 1024;
        };
};
```

iv. Then, edit /etc/bind/named.conf.local file field to point the "/var/lib/bind/db.lab". and add some additional lines, including the dnssec-policy <name> and inline-signing yes. (inline-signing automatically generates a db.lab.signed file...Maybe change and mess with this for RRSig records)

v. Restart bind on this machine.

vi. Run the command dig @127.0.0.1 lab DNSKEY +dnssec +short to just output the KSK for this zone. You will copy this key and paste it to the client machine. These will be carried into the next step.

d. On the client machine

i. Create a file to hold the trust anchor. I called mine lab-trust.key. I added it to “etc/bind/lab-trust.key”. In the file, I added what is seen below. As seen below, the 257 recognizes this key as being the KSK for the lab zone. The base 64 key should be pasted at the end, with this format: “base-64 key”;. The client must directly trust lab. as the top of the chain instead of a root server.

```
trust-anchors {
        lab. static-key 257 3 8 "AwEAAdyubupVYO97+QR5LxX3eIh8vxOI5bxfVz6ieZqMF19e
5Ul2SD+MyyY5iM1VRutYEpLmfbMDDuW3ZElJzN4BRP8huMuzRf8V4nAuB3poxIUs+kRywr4n6et46oCBF
7oWmcw792jAKKXFiuegwZ3s5NvrZufCckq5wzgaPvMxJLZLkalddzkc0KuvwAtmGsOTT7lkn3sx59JgWG
SYiL+3FNZFSZPlcDyw3QuIO1hN+Vq2NwuSR2LkOFX3miCKx4d07hr27TtP3LGJiBhcyMeJRT3i4bOBbCt
wGelA6xq5i4k1fg2+R2e+2VZtqk35GJsh8c0yqKoahz96ZCSrWCHm690=";
};
```

ii. Also, add the key file to your named.conf include path: <include_key>. This should be added near the other include paths in the file. See below:

```
include "/etc/bind/lab-trust.key";
```

iii. Then, to check the setup, dig @parent_ip <domain> A +dnssec. You should see and additional ad flag. If so,

e. On Child Server

i. Firstly, copy over db.lab file from /etc/bind/ to /var/lib/bind/. You will need to edit it like so:

```
$TTL    86400   ; 1 day
jmu.lab.        IN      SOA     ns.jmu.lab. root.jmu.lab. (
                             10         ; Serial
                          43200         ; Refresh
                            900         ; Retry
                        1814400         ; Expire
                           7200 )       ; minimum
;

jmu.lab.        IN      NS      ns.jmu.lab.

ns.jmu.lab.     IN      A       192.168.107.128
example IN      A       192.168.107.50
shea    IN      A       192.168.107.69
john    IN      A       192.168.107.67
alley   IN      A       192.168.107.68
king    IN      A       192.168.107.66
main    IN      A       192.168.107.65
quad    IN      A       192.168.107.64
```

ii. Then, in named.conf.local, edit /etc/bind/named.conf.local file field to point the "/var/lib/bind/db.lab". and add some additional lines, including the dnssec-policy <name> and inline-signing yes.

iii. Restart bind.

f. Back on the client

i. dig @child_ip <domain> A +dnssec. Yous should see an A record, RRSIG for A record (POSSIBLY KEYS?)

g. Back on Child Server

i. Generate the Delegation Signer on Child and give it to the parent to link them.

1. Run: dnssec-dsfromkey -2 Kjmu.lab.+008+<KEYID>.key that should produce something like: jmu.lab. IN DS 34939 8 2 <digest>

2. Then, copy this data and save it for the next step on the parent server.

h. On the Parent

i. In /var/lib/bind/db.lab, add the following line for the DS record:

```
$TTL    86400
lab.    IN      SOA     lab. root.lab. (
                              3         ; Serial
                          43200         ; Refresh
                            900         ; Retry
                        1814400         ; Expire
                           7200 )       ; minimum

lab.    IN      NS      ns.lab.
ns      IN      A       192.168.107.129

jmu     IN      NS      ns.jmu.lab.
ns.jmu  IN      A       192.168.107.128
; bc    IN      A       192.168.107.1
computer        IN      A       192.168.107.2
jmu IN DS 34939 8 2 9682051422A73107F882F98D04F47492ECA60648DC071428E43477AC7F1:
42C2
```

i. EXTRA

   i. To get DNSKEY and RRSIG files, I ran a dig @server_address DNSKEY +dnssec

   ii. Running sudo rndc flush is not a bad idea to run if you run into errors with keys, especially if you end up regenerating keys.

Source: Ubuntu Server Documentation - https://documentation.ubuntu.com/server/how-to/networking/install-dns/ Date accessed: 1/20/2026

Bind9 Man Pages, Date accessed 1/20/2026

https://bind9.readthedocs.io/en/v9.20.17/chapter3.html

Resolvctl Man Pages, Date accessed: 1/26/2026

https://www.freedesktop.org/software/systemd/man/latest/resolvectl.html

Resolved Man Pages, Date accessed: 1/27/2026

https://www.freedesktop.org/software/systemd/man/latest/systemd-resolved.service.html